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2025

(Session : 2024-27)

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Sections as directed.

Section – A

(Multiple-Choice Questions)

(Compulsory)

1. Choose the correct answer of the following: 2×10 = 20

(a) Which of the following is not a feature of Java ?

(a) Platform Independent

(b) Object oriented

(c) Pointer-based memory Access

(d) Robust

(b) Which keyword is used to create an object in Java ?

(a) create (b) new

(c) object (d) init

(c) Which of the following is a valid Java data type ?

(a) number (b) real

(c) decimal (d) float

(d) Which method is the entry point of a Java program ?

(a) start() (b) execute()

(c) main() (d) run()

(e) Which OOP principle allows multiple methods with the same name but different parameters ?

(a) Inheritance

(b) Polymorphism

(c) Encapsulation

(d) Abstraction

(f) What is the default value of an int variable in Java ?

(a) 1 (b) 0

(c) null (d) -1

(g) Which class is the parent of all Java classes ?

(a) Object (b) Main

(c) Class (d) JavaRoot

(h) Which keyword is used to inherit a class ?

(a) implement (b) inherit

(c) extends (d) superclass

(i) Applets in Java run inside a :

(a) Browser (b) Complier

(c) File Manager (d) Console

(j) Who invented Java Programming?

(a) Guido van Rossum

(b) James Gosling

(c) Dennis Ritchie

(d) Bjarne Stroustrup

Section – B

(Short-answer Type Questions)

Answer any **four** questions of the following : 5×4 = 20

2. What is an applet ? Explain its life cycle methods. Explain JSP life cycle with a neat diagram.
3. Explain Java methods : scope, defining, calling, and returning values.
4. Explain constructors and their types with examples.

5. Discuss the concept of polymorphism in Java with runtime and compile-time examples.

6. What is inheritance and define its type in Java ?

Section – C

(Long-answer Type Questions)

Answer any **two** questions of the following : 15×2 = 30

7. What is Object-Oriented Programming in Java ? Explain all OOP principles— Encapsulation, Inhreitance, Polymorphism, Abstracation— with well-structured examples.

8. Explain the architecture of Java in detail. Discuss the role of JVM, JRE, AND JDK.

9. Write short notes on the following :
   (a) Data Types in Java
   (b) Interfaces in Java
   (c) TCP/IP Networking in Java
   (d) Abstract Classes

10. Explain arrays, strings and I/O streams in Java with suitable examples.